Model Performance Plots

Pratham Shah

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1 XGBoost

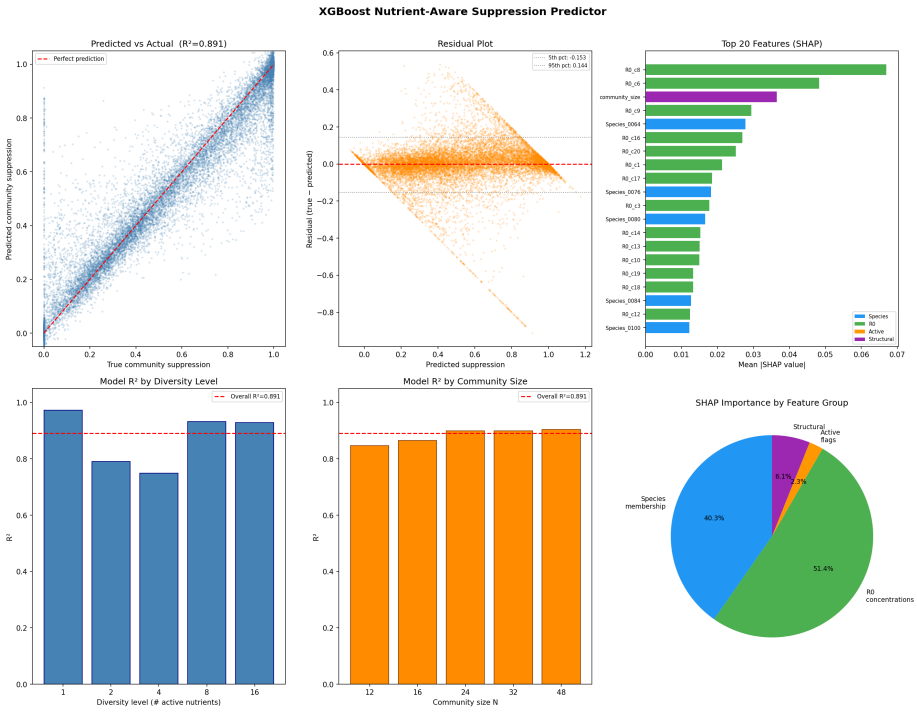


Figure 1: V1

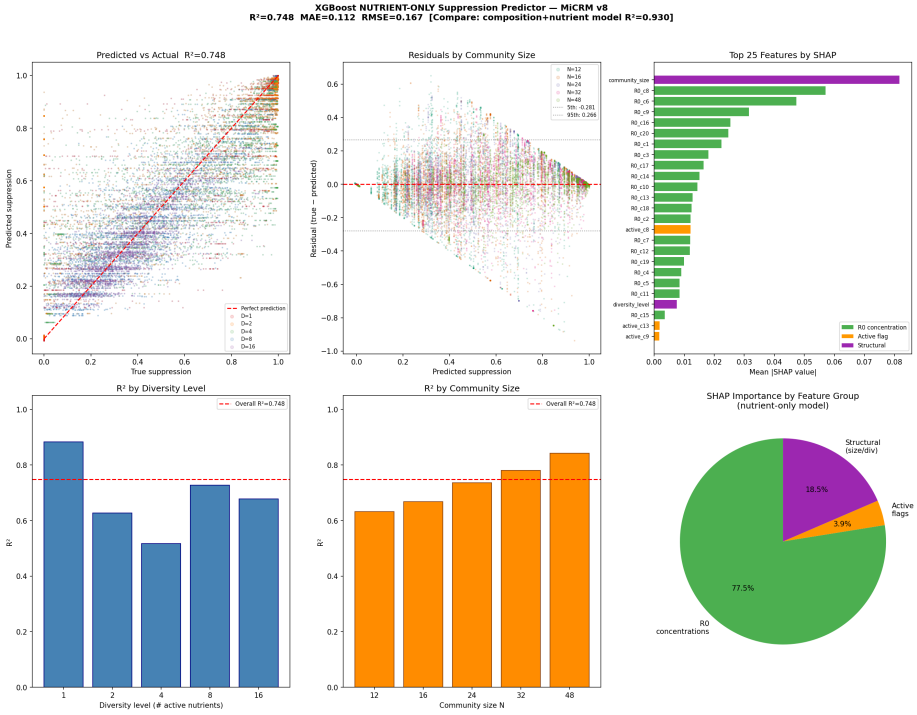

XGBoost NUTRIENT-ONLY Suppression Predictor — MiCRM v8
R²=0.748 MAE=0.112 RMSE=0.167 [Compare: composition+nutrient model R²=0.930]
Predicted vs Actual R²=0.748
True suppression
Predicted suppression
Perfect prediction
D=1
D=2
D=4
D=8
D=16
Residuals by Community Size
Residual (true − predicted)
N=12
N=16
N=24
N=32
N=48
50th: -0.281
95th: 0.266
Top 25 Features by SHAP
Mean |SHAP value|
R0 concentration
Active flag
Structural
R² by Diversity Level
Overall R²=0.748
Diversity level (# active nutrients)
R² by Community Size
Community size N
SHAP Importance by Feature Group
(nutrient-only model)
Structural
(size/div)
18.5%
Active
flags
3.9%
77.5%
R0
concentrations

Figure 2: V2

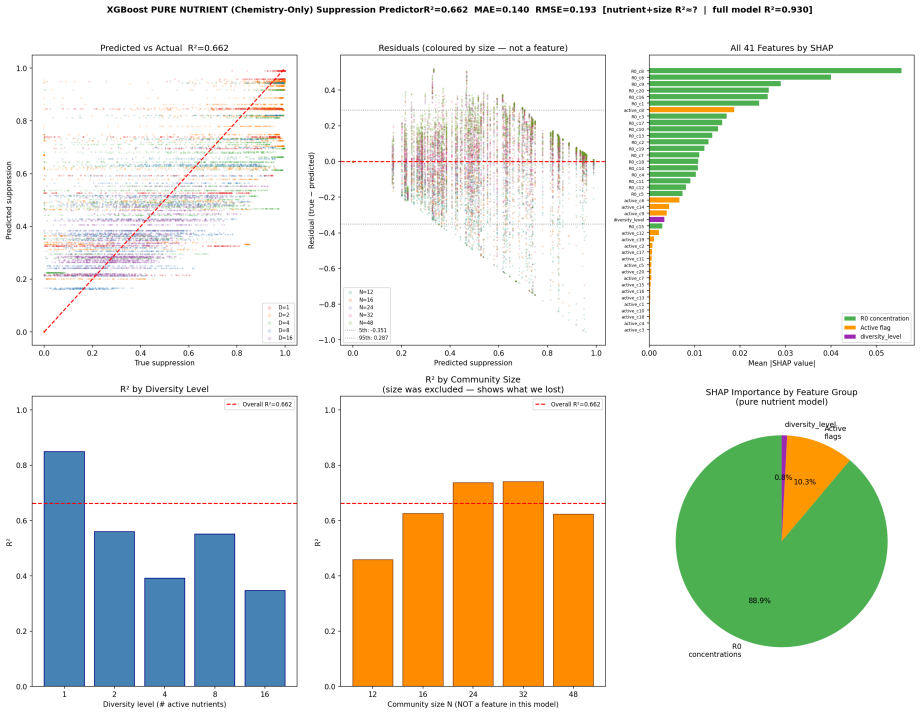

XGBoost PURE NUTRIENT (Chemistry-Only) Suppression PredictorR²=0.662 MAE=0.140 RMSE=0.193 [nutrient+size R²≈? | full model R²=0.930]
Predicted vs Actual R²=0.662
Predicted suppression
True suppression
D=1
D=2
D=4
D=8
D=16
Residuals (coloured by size — not a feature)
Residual (true − predicted)
Predicted suppression
N=12
N=16
N=24
N=32
N=48
50th: -0.351
95th: 0.287
All 41 Features by SHAP
Mean |SHAP value|
R0 concentration
Active flag
diversity_level
R² by Diversity Level
R²
Diversity level (# active nutrients)
Overall R²=0.662
R² by Community Size
(size was excluded — shows what we lost)
Community size N (NOT a feature in this model)
SHAP Importance by Feature Group
(pure nutrient model)
diversity_level
Active flags
0.8%
10.3%
88.9%
R0 concentrations

Figure 3: V3

2 LightGBM

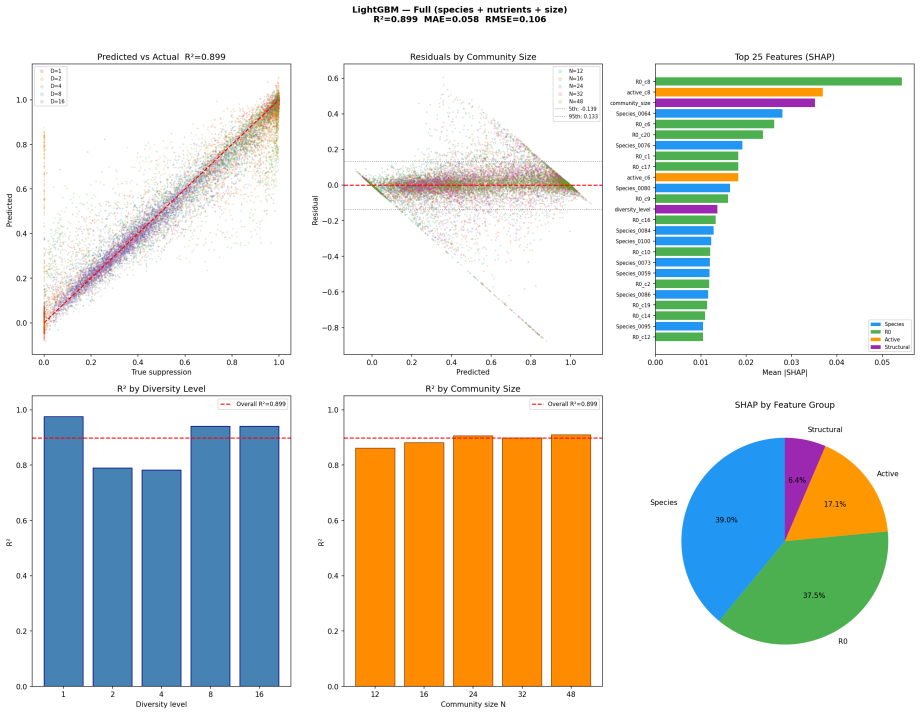


Figure 4: V1


LightGBM — Nutrients + size (no species)
R²=0.748 MAE=0.113 RMSE=0.167
Predicted vs Actual R²=0.748
True suppression
Predicted
D=1
D=2
D=4
D=8
D=16
Residuals by Community Size
Residual
N=12
N=16
N=24
N=32
N=48
5th: -0.272
95th: 0.269
Top 25 Features (SHAP)
Mean |SHAP|
community_size
R0_c8
R0_c6
active_c8
R0_c16
R0_c20
R0_c9
R0_c17
R0_c1
active_c20
R0_c10
diversity_level
R0_c14
R0_c2
R0_c7
active_c6
R0_c15
active_c9
R0_c12
R0_c4
R0_c3
R0_c11
R0_c19
R0_c18
active_c19
Species
R0
Active
Structural
R² by Diversity Level
Overall R²=0.748
R²
Diversity level
R² by Community Size
Community size N
SHAP by Feature Group
Structural
19.2%
59.2%
21.6%
R0
Active

Figure 5: V2

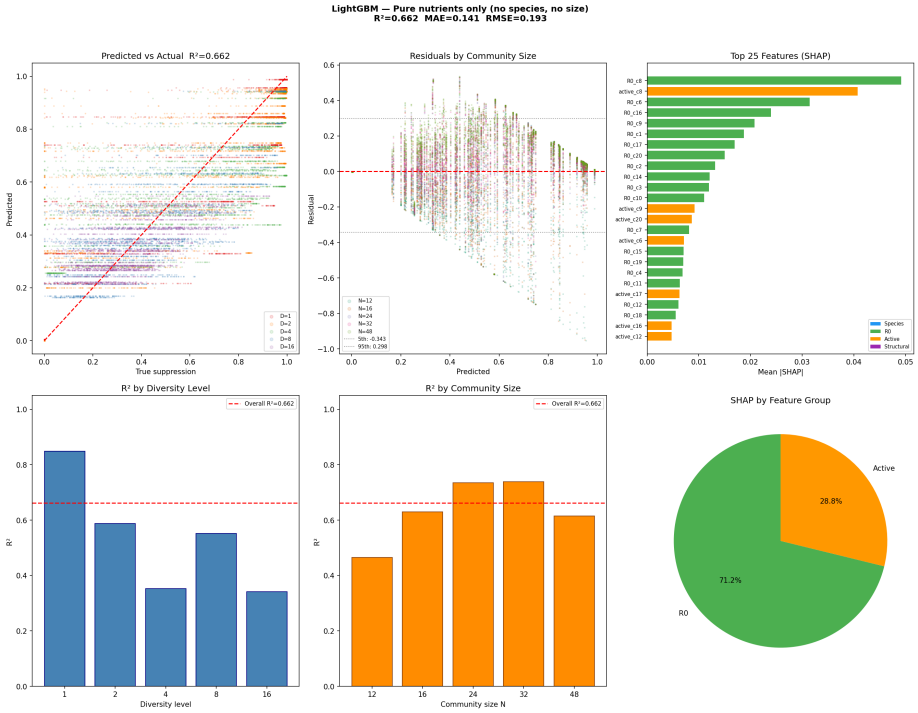

LightGBM — Pure nutrients only (no species, no size)
R²=0.662 MAE=0.141 RMSE=0.193
Predicted vs Actual R²=0.662
True suppression
Predicted
D=1
D=2
D=4
D=8
D=16
Residuals by Community Size
Residual
Predicted
N=12
N=16
N=24
N=32
N=48
50th: -0.343
95th: 0.298
Top 25 Features (SHAP)
Mean |SHAP|
Species
R0
Active
Structural
R² by Diversity Level
Overall R²=0.662
R²
Diversity level
R² by Community Size
Overall R²=0.662
R²
Community size N
SHAP by Feature Group
Active
28.8%
71.2%
R0

Figure 6: V3

3 CatBoost

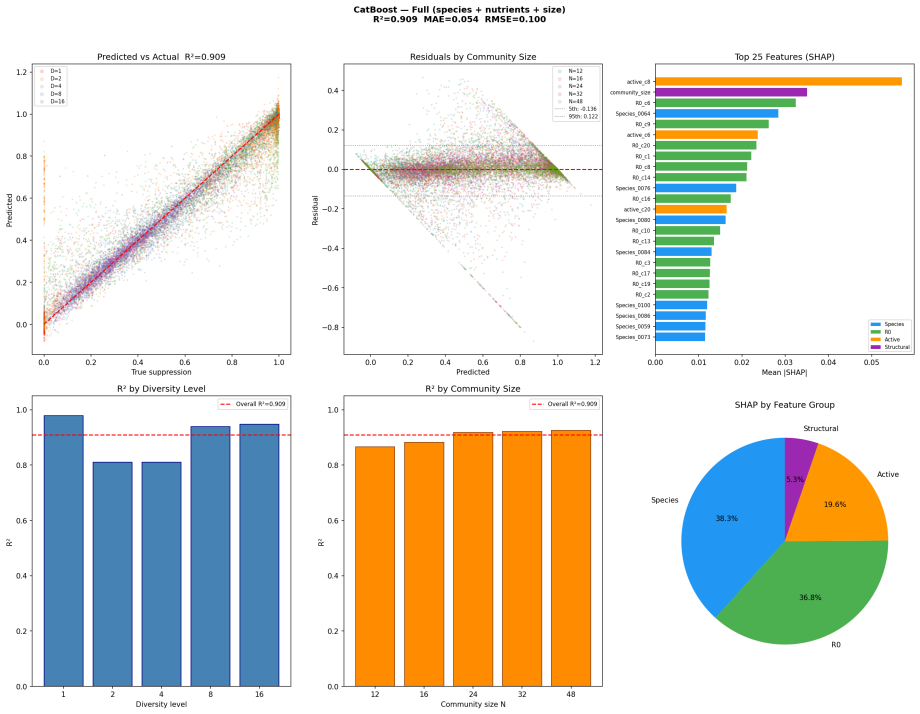


Figure 7: V1


CatBoost — Nutrients + size (no species)
R²=0.758 MAE=0.111 RMSE=0.163
Predicted vs Actual R²=0.758
True suppression
Predicted
D=1
D=2
D=4
D=8
D=16
Residuals by Community Size
Residual
Predicted
N=12
N=16
N=24
N=32
N=48
50%: 0.288
95th: 0.247
Top 25 Features (SHAP)
Mean |SHAP|
community_size
R0_c8
active_c8
active_c6
R0_c6
R0_c20
R0_c9
R0_c1
R0_c16
active_c20
active_c14
R0_c17
R0_c10
R0_c3
R0_c2
active_c19
R0_c7
R0_c19
diversity_level
R0_c14
R0_c12
active_c12
R0_c4
R0_c18
active_c13
Species
R0
Active
Structural
R² by Diversity Level
Overall R²=0.758
R²
Diversity level
R² by Community Size
Overall R²=0.758
R²
Community size N
SHAP by Feature Group
Structural
17.8%
R0
52.4%
29.9%
Active

Figure 8: V2

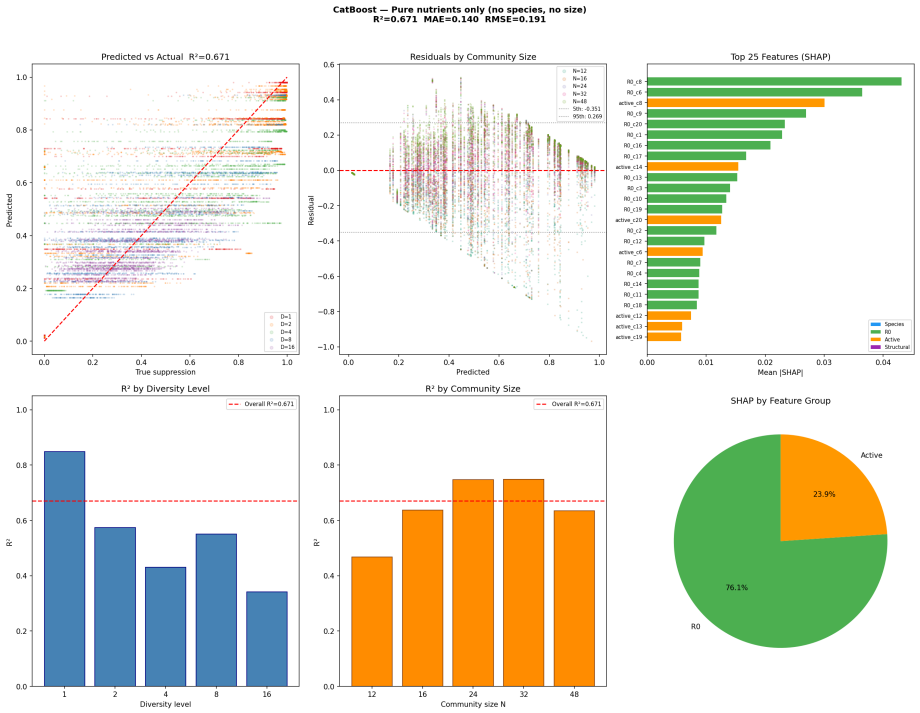

CatBoost — Pure nutrients only (no species, no size)
R²=0.671 MAE=0.140 RMSE=0.191
Predicted vs Actual R²=0.671
True suppression
Predicted
D=1
D=2
D=4
D=8
D=16
Residuals by Community Size
Predicted
Residual
N=12
N=16
N=24
N=32
N=48
50%: -0.161
90%: 0.269
Top 25 Features (SHAP)
Mean |SHAP|
Species
R0
Active
Structural
R² by Diversity Level
Overall R²=0.671
Diversity level
R²
R² by Community Size
Overall R²=0.671
Community size N
R²
SHAP by Feature Group
Active
23.9%
76.1%
R0

Figure 9: V3

4 TabPFN

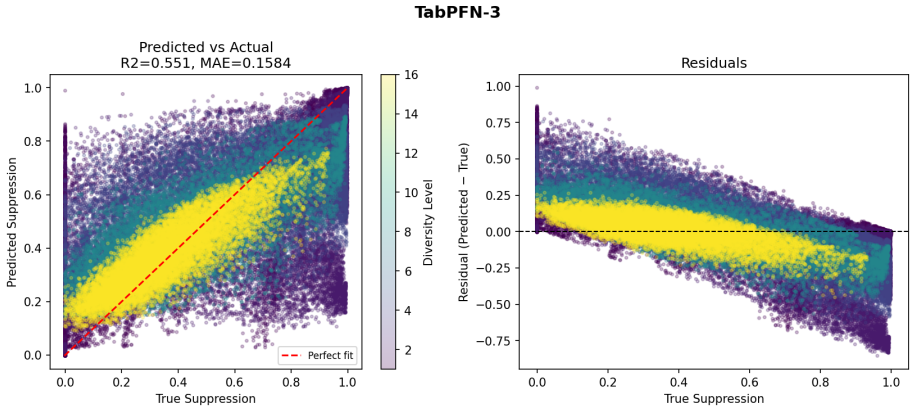


Figure 10: V1-Diagnostics

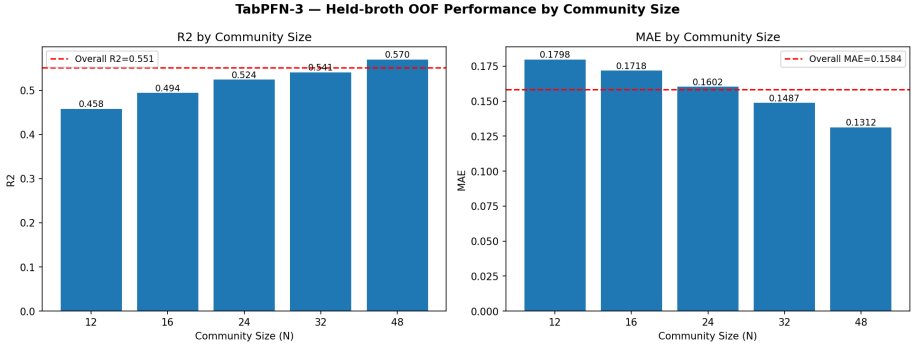


Figure 11: V1-Community Size Breakdown

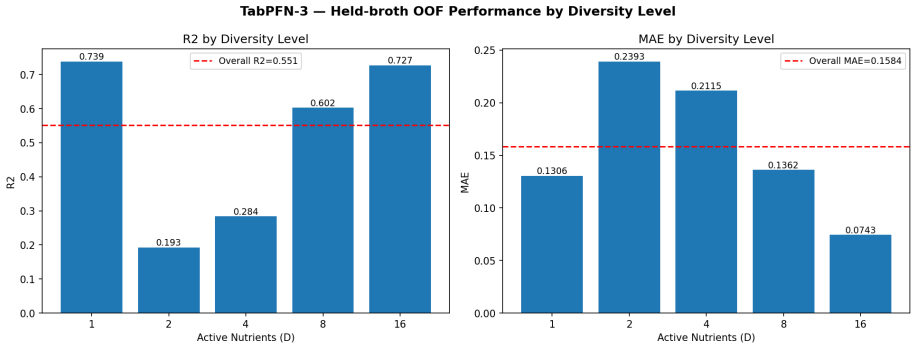


Figure 12: V1-Diversity Level Breakdown

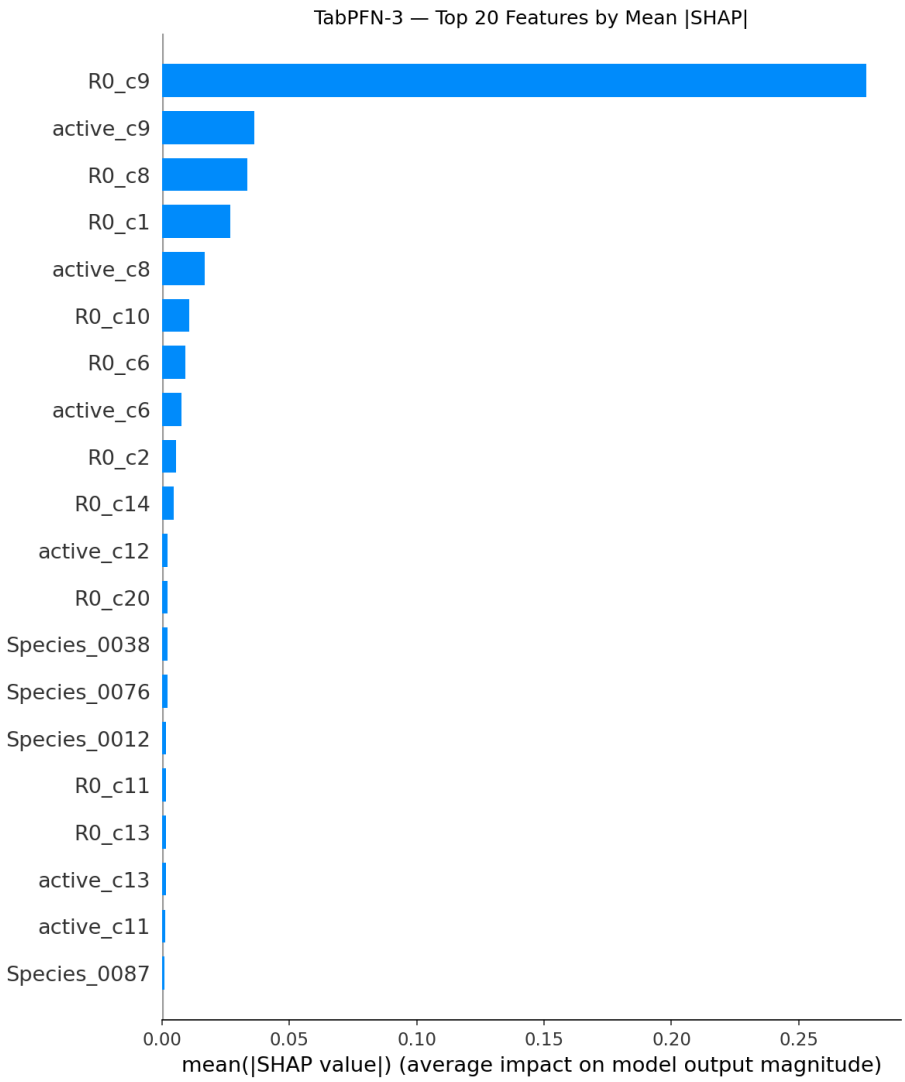


Figure 13: V1-SHAP

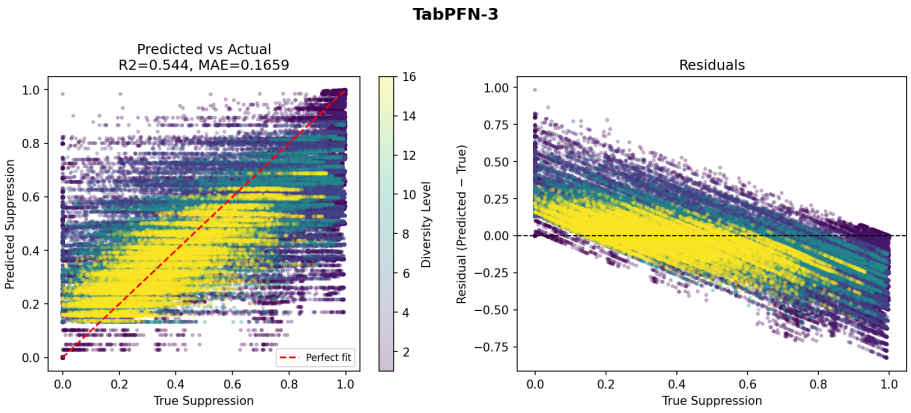


Figure 14: V2-Diagnostics

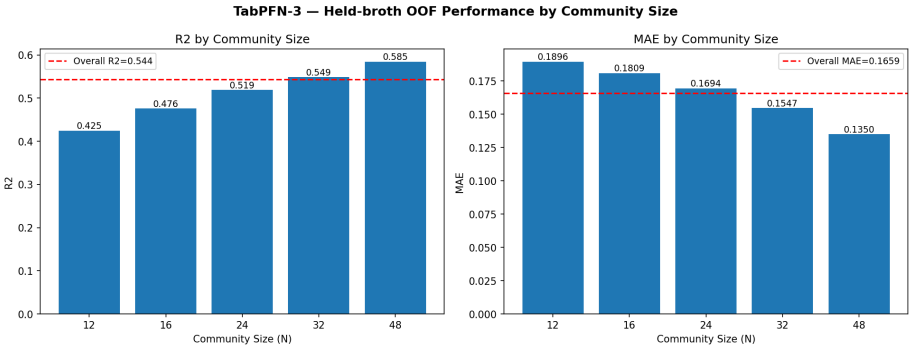


Figure 15: V2-Community Size Breakdown

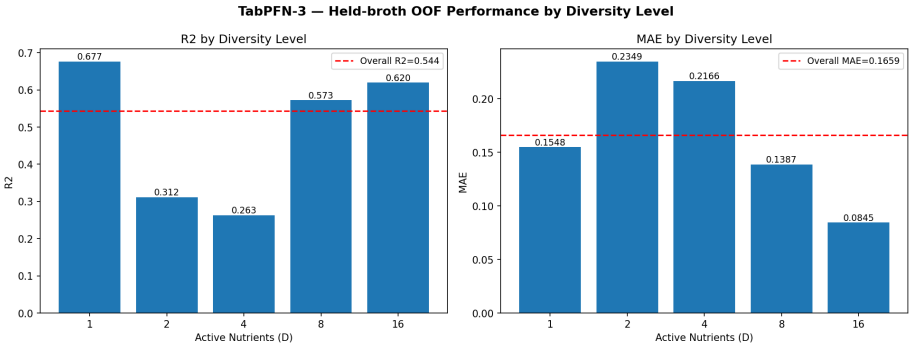


Figure 16: V2-Diversity Level Breakdown

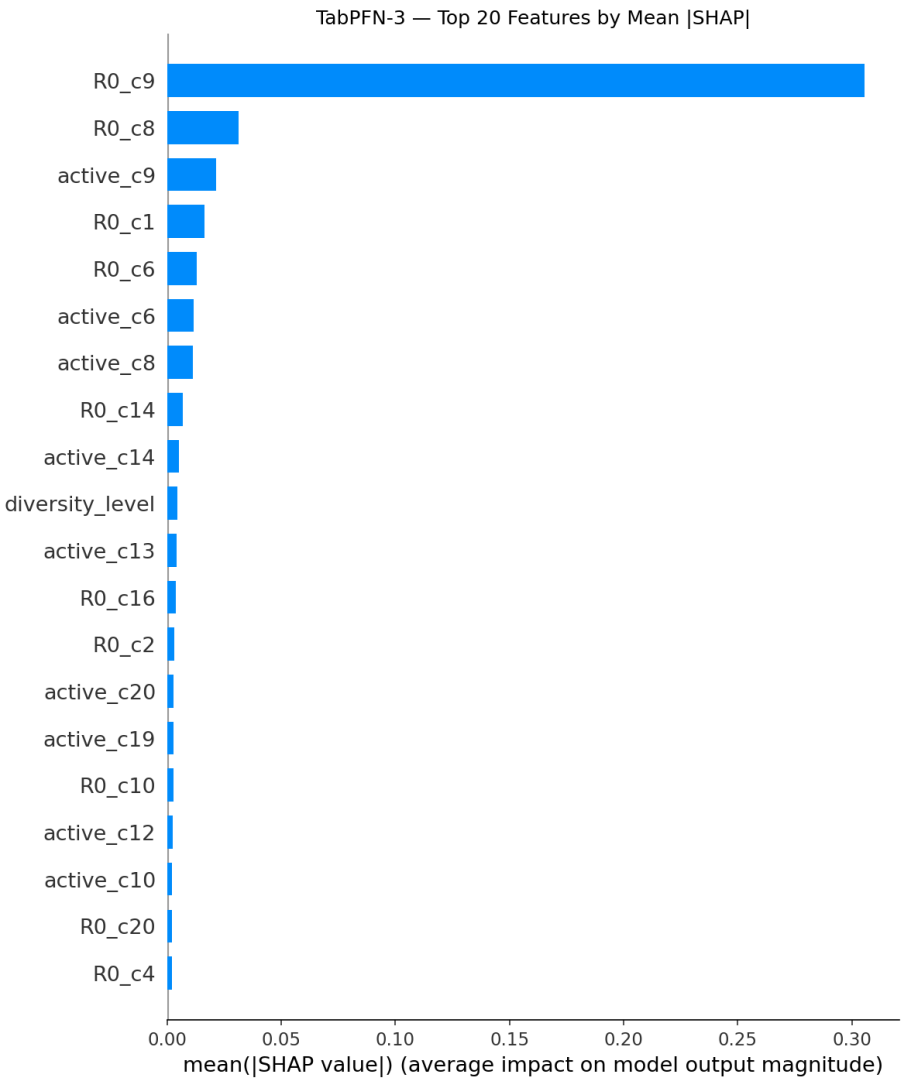


Figure 17: V2-SHAP

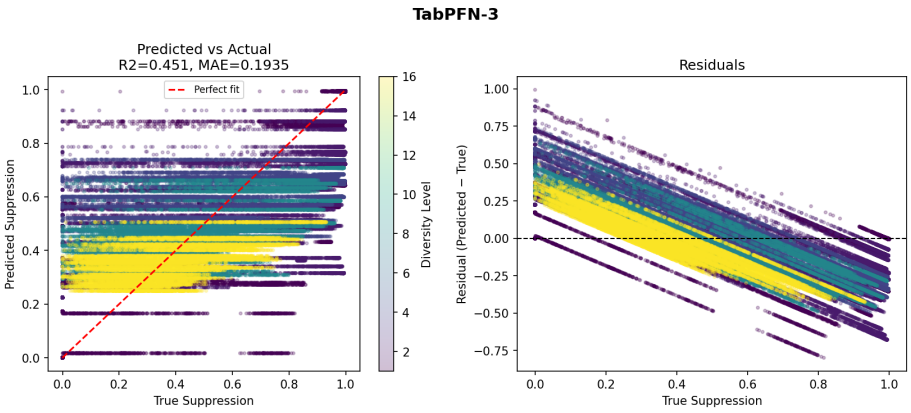


Figure 18: V3-Diagnostics

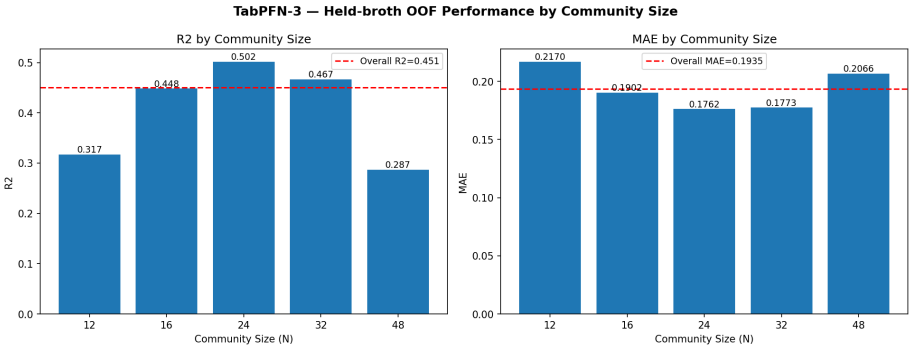


Figure 19: V3-Community Size Breakdown

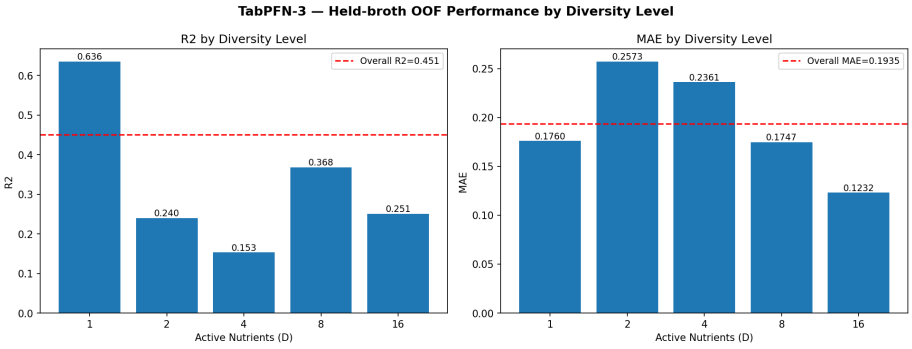


Figure 20: V3-Diversity Level Breakdown

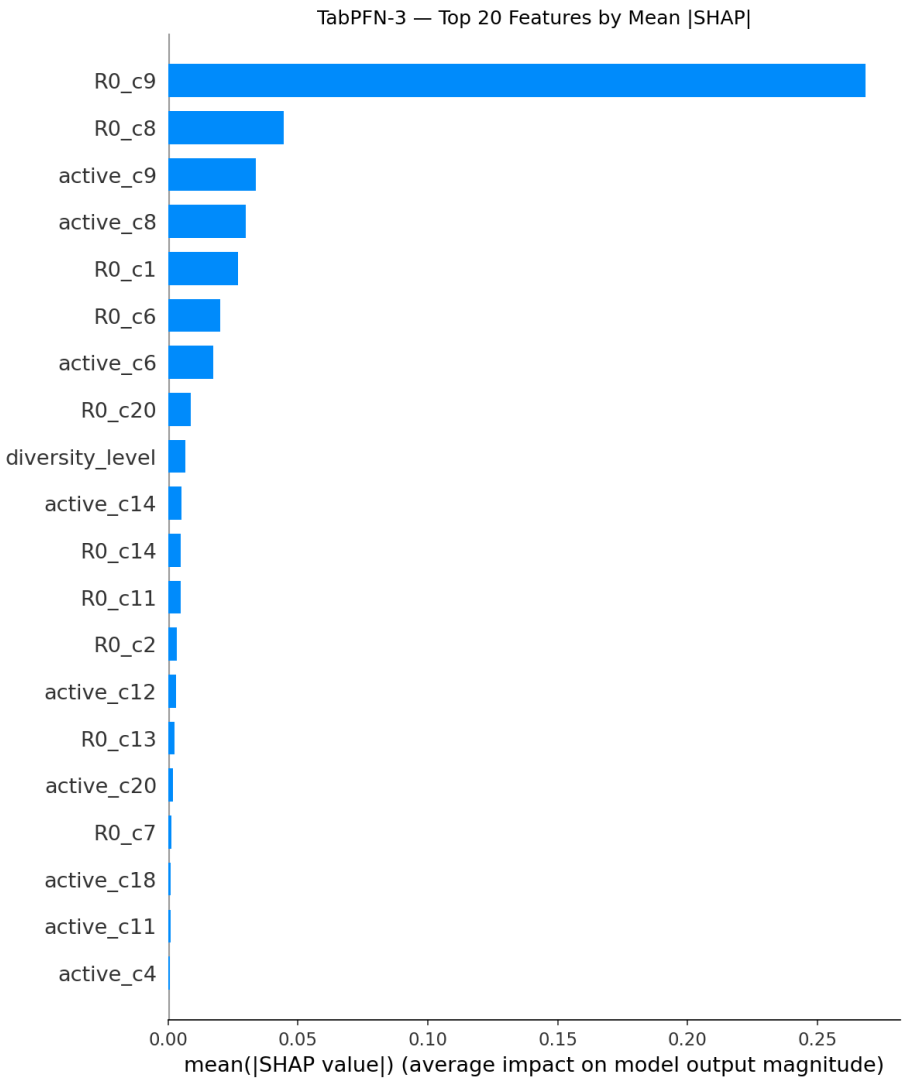


Figure 21: V3-SHAP